St. Philomena's College (Autonomous), Mysore			
PG Department of Biochemistry			
Question Bank (Revised Curriculum 2020-21 onwards)			
First Year- First Semester (2020-21 Batch)			
Course Title (Paper Title): Human Physiology (SC) QP Code - 84131			
Sl. No	**Unit**	**Questions**	**Marks**
1	1	What is differential counting? Give its significance.	2
2	1	Which are the different types of tissues? Give their biological significance.	2
3	1	How serum does differ from that of plasma?	2
4	1	Which are the haemopoietic organs? Give their functions.	2
5	1	What are anticoagulants? Give an example.	2
6	1	What is Rh factor?	2
7	1	What are the different types of muscles?	2
8	1	Define haemostasis.	2
9	1	What is plasma? Give its composition.	2
10	1	What are plasma proteins? Give an example.	2
11	1	Name the different types of plasma proteins.	2
12	1	How do you digest a clot?	2
13	1	Define homeostasis.	2
14	1	What are anti coagulants? Give an example.	2
15	1	Mention the role of WBC as phagocyte cell.	2
16	1	Give examples for anti-coagulants.	2
17	1	What is coagulation?	2

18	1	Distinguish between plasma & serum.	2
19	1	Enlist the functions of CSF.	2
20	1	Enlist the functions of Lymph.	2
21	1	Define CSF and its function.	2
22	1	How CSF is formed?	2
23	1	Give the functions of blood.	2
24	1	State the percentage composition of blood.	2
25	1	What are platelets?	2
26	1	Classify the lymphocytes of blood and mention its key function.	2
27	1	Write the normal value of RBC in male & female.	2
28	1	Define blood pressure and hypertension.	2
29	1	Write the normal value and life span of leukocytes.	2
30	1	Write the normal value and life span of erythrocytes.	2
31	1	What is a buffer? List the three intracellular buffers.	2
32	2	Draw the cross section of the human heart and label the parts	2
33	2	What is ECG? Why is it used?	2
34	2	What is meant by Vital Capacity?	2
35	2	Differentiate between arteries and veins.	2
36	2	Define heart rate and stroke volume.	2
37	2	What are the events of cardiac cycle?	2
38	2	Define ECG.	2

39	2	What is chloride shift?	2
40	2	What is Einthoven's triangle? Mention its role.	2
41	2	Draw the Einthoven's triangle	2
42	2	Define tidal volume.	2
43	2	What is the normal blood pressure? How do you find it?	2
44	2	Define cardiac output.	2
45	2	Why are the right and left lungs slightly different in size and shape?	2
46	2	What are the difference between an atrium and ventricle?	2
47	2	Which lung volume cannot be measured directly and why?	2
48	2	How lung volume is related to lung capacity?	2
49	2	Define respiration.	2
50	2	Enlist upper and lower respiratory organs.	2
51	2	Define tidal volume and residual volume.	2
52	2	Enlist the different parts of respiratory system.	2
53	2	What are cardiac muscles? Give its structure.	2
54	2	What are the events involved in cardiac cycle?	2
55	3	What is GFR? Give its significance.	2
56	3	Give the major functions of liver.	2
57	3	List out the composition of urine.	2
58	3	What is GFR?	2
59	3	Define Glomerulus Filtration Rate.	2

60	3	What is a bowman's capsule? Give its importance.	2
61	3	What is glomerus? Give its importance.	2
62	3	Mention the secretary function of liver.	2
63	3	Enlist the excretory function of liver.	2
64	3	What are kupffer cells? Give their importance.	2
65	3	What are Kupffer cells? Where are they present?	2
66	3	What are podocytes? Give its significance.	2
67	3	What are endothelial cells? Give its importance.	2
68	3	What is bile?	2
69	3	What is bile? Where does it produced & stored?	2
70	3	What are nephrons? Mention its role.	2
71	3	Give any two functions of bile.	2
72	3	Give the composition of Bile.	2
73	3	Mention the functions of kidney.	2
74	3	Which vessels carry and drain blood from kidneys?	2
75	3	By what mechanisms tubular reabsorption of substances takes place?	2
76	4	What are the enzymes involved in the digestion of proteins?	2
77	4	What are the enzymes involved in the digestion of carbohydrates?	2
78	4	What are the enzymes involved in the digestion of fats?	2
79	4	How does blocking of H2 receptor stop acid production in the stomach?	2
80	4	Mention the role of pancreas in digestion of food.	2

81	4	Name the hormones secreted by GI tract.	2
82	4	Enlist the hormones secreted by GIT.	2
83	4	What are villi? Give their role.	2
84	4	What are the secretions of gastrointestinal system?	2
85	4	Write the functions of gall bladder.	2
86	4	Define absorption and digestion.	2
87	4	Name the four layers of the gastro intestinal tract.	2
88	4	Name any two GIT hormones and mention its functions.	2
89	5	What are the different parts of brain?	2
90	5	How right brain does differ from left brain?	2
91	5	Define action potential.	2
92	5	Define resting potential.	2
93	5	Enlist the components of nervous system.	2
94	5	What is synapse?	2
95	5	Write the functions of cerebellum.	2
96	5	What is reflex action?	2
97	5	Why axon terminals are sometimes called “Biological transducers”.	2
98	5	What is myelin sheath? Give its importance.	2
99	5	What are neurotransmitters? Give an example.	2
100	5	Draw the neuromuscular junction.	2
101	5	Give the classification of peripheral nervous system.	2

102	5	What are the coverings of skeletal muscle?	2
103	5	Name the proteins present in the muscle?	2
104	5	Give the structure of smooth muscle.	2
105	5	Name the different types of muscle.	2
106	5	What are troponins? Mention their significance.	2
107	5	What are Glial cells? Give its significance.	2
108	5	Name the two Glial cell types that form myelin sheath.	2
109	6	List the four major types of ion channels found in neuron.	2
110	6	What do you mean by IVF?	2
111	6	What do you mean by Invitro Fertilization?	2
112	6	What is ovulation?	2
113	6	Enlist the parts of female reproductive system.	2
114	6	Enlist the parts of male reproductive system.	2
115	6	Define HCG.	2
116	6	Write the functions of ovaries.	2
117	6	Mention the hormones in lactation.	2
118	6	Write the role of oxytocin.	2
119	6	What abnormalities in the uterus could cause infertility?	2
120	6	Enlist the factors influencing ovulatory cycle.	2
121	6	Define oogenesis.	2
122	6	What is menstrual cycle?	2

1	1	What is blood? Enumerate various components and functions of blood.	5
2	1	What are plasma proteins? Give their classification with functions.	5
3	1	Classify WBCs and explain the structure and function of each type.	5
4	1	Write short note on WBCs.	5
5	1	Discuss the structural features and functions of WBC.	5
6	1	Classify WBCs and discuss functions of each type.	5
7	1	What are the different types of leucocytes? Give their structure with function.	5
8	1	What is the significance of high and low White blood cell counts? explain	5
9	1	What is a blood group? Classify blood group.	5
10	1	What is a blood group? Explain ABO blood group system in detail.	5
11	1	Describe the composition and functions of lymph.	5
12	1	Describe the composition and functions of CSF.	5
13	1	Write a note on Cerebrospinal Fluid.	5
14	1	Describe blood clotting mechanism.	5
15	1	Explain the mechanism of blood clotting.	5
16	1	Explain in detail the mechanism of blood coagulation process.	5
17	2	Enlist conducting elements of heart and explain conduction system of heart.	5
18	2	Describe the structure of heart and give its functions.	5
19	2	Describe the internal structure of heart.	5
20	2	What are the unique properties of cardiac muscle essential for its function? Explain in detail.	5

21	2	Write short note on conducting system of heart with cardiac cycle.	5
22	2	Define blood pressure and explain factors affecting blood pressure.	5
23	2	Write short note on blood pressure.	5
24	2	Explain the regulation of blood pressure.	5
25	2	Discuss the physiological variations and regulation of blood volume.	5
26	2	Explain cardiac cycle.	5
27	2	What are the events of cardiac cycle? Explain.	5
28	2	Give a brief account on ECG.	5
29	2	Explain ECG. Mention the different waves of ECG.	5
30	2	How baroreceptors play an important role in the control of blood pressure?	5
31	2	Give the structure and functions of lungs.	5
32	2	Draw and explain the different parts of a respiratory system.	5
33	2	Explain the physiology of respiration.	5
34	2	Explain the mechanism of respiration.	5
35	2	What is tidal volume? Explain the mechanism of respiration.	5
36	2	Explain the role of lungs in maintain acid base balance.	5
37	2	Explain about exchange of gases between blood and tissues.	5
38	3	Describe the process of urine formation.	5
39	3	Discuss the process of urine formation.	5
40	3	List and explain the six major functions of kidney.	5
41	3	Describe the physiology of excretory system.	5

42	3	Describe the structure of nephron.	5
43	3	How Glomerular filtrate is is formed? Explain.	5
44	3	Discuss on tubular reabsorption step of urine formation.	5
45	3	Discuss the process of selective reabsorption in the formation of urine.	5
46	3	Explain the renal regulation of acid-base balance.	5
47	3	Enlist the functions of liver. Explain any two of them in detail.	5
48	3	Explain the secretory & excretory functions of liver.	5
49	4	Draw a neat labelled diagram of digestive system and explain their functions.	5
50	4	Enlist the organs involved in digestive system and give functions of each organ.	5
51	4	Explain the functions of the digestive system.	5
52	4	How are fats digested and absorbed?	5
53	4	How are proteins digested and absorbed?	5
54	4	How are carbohydrates digested and absorbed?	5
55	4	Discuss the digestion and absorption of nutrients from small intestine.	5
56	4	Write a note on pancreatic digestion.	5
57	4	Explain the role of pancreatic enzymes in digestion.	5
58	4	Explain the secretion of HCl in the gastric juice. List the factors regulating HCl secretion.	5
59	4	Describe the phases and regulation of gastric secretion.	5
60	4	Write a note on phases of gastric secretion.	5
61	5	Explain the ultra structure of muscle.	5

62	5	Write the structure and functions of actin and myosin.	5
63	5	Explain the chemical changes that take place during muscle contraction.	5
64	5	Explain the steps of muscle contraction in detail.	5
65	5	Explain the physiology of muscle contraction with suitable diagram.	5
66	5	Explain the structure of skeletal muscle and physiology of muscular contraction.	5
67	5	Write a note on muscle contraction.	5
68	5	Write short note on action potential & neuromuscular junction.	5
69	5	What are the different parts of brain? Give their functions.	5
70	5	Define and discuss about action potential.	5
71	5	Explain the process of transmission of nerve impulses.	5
72	6	Describe the principal events of oogenesis.	5
73	6	Write a note on ovarian cycle.	5
74	6	Write about natural methods of contraception.	5
75	6	Write an essay on menstrual cycle.	5
76	6	Give an account of the chemical methods of female contraception.	5
77	6	Enlist and explain any three Sexually Transmitted Diseases.	5
1	1	Enumerate the blood clotting factors and describe the biochemistry of the coagulation of blood.	10
2	1	Enumerate the blood clotting factors and explain the blood coagulation cascade.	10
3	1	Define coagulation. Describe in detail mechanism of blood coagulation.	10

4	1	What is coagulation? Discuss in detail about intrinsic and extrinsic pathway of blood coagulation.	10
5	1	Explain the blood clotting mechanism.	10
6	1	Explain blood clotting cascade.	10
7	1	Explain in detail the mechanism of blood clotting.	10
8	1	Enlist clotting factors. Explain in detail blood clotting mechanism.	10
9	2	Draw a neat labelled diagram of conduction system of heart. Explain in detail the cardiac cycle.	10
10	2	Describe the interior structure of heart. Explain cardiac cycle in detail.	10
11	2	Explain cardiac cycle with normal ECG.	10
12	2	Hat is electrocardiogram? Explain its clinical variation in diseases.	10
13	2	What is ECG? Explain in detail.	10
14	2	What is cardiac cycle? Explain the various events of cardiac cycle.	10
15	2	Define blood pressure. Discuss in detail short term & long term mechanism involved in regulation of blood pressure.	10
16	2	Explain the following: Blood pressure & cardiac output	10
17	2	Describe the factors which maintain the blood pressure.	10
18	2	Define blood pressure. Discuss the factors affecting & regulation of Blood pressure.	10
19	2	Describe the structure of lungs. Explain the respiratory regulation of acid base balance.	10
20	2	Describe the structure of lungs. Explain in detail the transport of O_2 & CO_2 during respiration.	10
21	2	Discuss the structure of alveoli and explain the process of exchange of gases.	10

22	2	Describe the location & gross anatomy of lungs. Explain exchange of gases at alveolar and cellular level.	10
23	2	Enlist different organs involved in respiration. Explain the mechanism of respiration and exchange of gases at lung and tissue level.	10
24	2	Write in detail physiology of respiration and transport of O_2 & CO_2.	10
25	2	Explain the mechanism of respiration.	10
26	2	Discuss the mechanism of respiration.	10
27	2	Explain the mechanism of respiration.	10
28	2	Explain the regulation of respiration.	10
29	2	Explain the physiology of respiration.	10
30	2	What is respiration? Discuss in detail about mechanism and regulation of respiration.	10
31	2	Describe the anatomy of lungs. Discuss the mechanism of respiration.	10
32	2	Describe the exchange of gases in the lungs.	10
33	3	Describe the structure of liver. Explain the function of liver in detail.	10
34	3	What is bile? How is it formed? Give its composition and functions.	10
35	3	Describe the structure of kidney. Explain the role of kidneys in the maintenance of acid-base balance.	10
36	3	Explain the role of lungs and kidneys in maintenenance of acid-base balance.	10
37	3	Describe the structure of nephron .explain in detail the physiology of urine formation.	10
38	3	Discuss the anatomy of kidney and explain Glomerular filtration in detail.	10
39	3	Explain the formation of urine. How the acid-base balance is regulated?	10

40	3	How is urine formed? Explain the mechanism.	10
41	3	Explain the formation and composition of urine.	10
42	3	Explain the formation of urine.	10
43	3	Write short note on Glomerular filtration and tubular secretion.	10
44	4	Describe the anatomy of stomach. Explain the production of HCl.	10
45	4	Discuss the anatomy of stomach. Explain the phases of gastric secretion.	10
46	4	Explain in detail the absorption of various nutrients from digestive tract.	10
47	4	Describe the structure of a villus and its function in absorption of nutrients.	10
48	4	List out the major digestive hormones & describe their function in the process of digestion.	10
49	4	Explain detail the digestion, transportation and absorption of carbohydrates.	10
50	4	Explain detail the digestion, transportation and absorption of fats.	10
51	4	Explain detail the digestion, transportation and absorption of proteins.	10
52	5	Give the structure of brain and state its functions.	10
53	5	Draw and explain the structure of neuron.	10
54	5	Describe the structure of neuron. Explain the features of the nerve impulse.	10
55	5	Describe the different phases of action potential.	10
56	5	Explain the conduction of nerve impulse.	10
57	5	What are neurotransmitters? Give their types & explain their functions.	10
58	5	Explain the ultra structure of muscle and explain its contraction.	10

59	5	Discuss anatomy of skeletal muscle and explain in detail mechanism of muscle contraction.	10
60	5	Write a note on mechanism of muscle contraction.	10
61	5	Explain the mechanism of muscle contraction.	10
62	5	Describe in detail molecular mechanism of skeletal muscle contraction.	10
63	5	Explain the different types of muscle proteins. Discuss on role of calcium & ATP in muscle contraction.	10
64	5	Explain the role of actin, myosin and calcium in muscle contraction.	10
65	5	Describe various types of muscle and give the mechanism of muscle contraction.	10
66	6	Explain the anatomy and physiology of female reproductive system	10
67	6	Describe the female reproductive system and explain its function.	10
68	6	Describe the male reproductive system and explain its function.	10
69	6	Explain the process of pregnancy.	10
70	6	Explain various types of birth control methods and compare their effectiveness.	10
71	6	Explain the various phases of the ovarian cycle.	10
72	6	Write about the natural methods of contraception. Add a note on IVF.	10
73	6	Enlist and explain any two Sexually Transmitted Diseases in detail.	10
74	6	Enlist various organs of female reproductive system and explain ovulation cycle.	10

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Model Question Paper			
St. Philomena's College (Autonomous), Mysore			
I Semester M.Sc-Final Examination			
Subject: Biochemistry			
Title: Human Physiology (SC)			
Time: 3 Hours			**Max Marks: 70**
PART-A			
Answer any TEN of the following:			**10x2=20**
1.		What are anti coagulants? Give an example.	**2**
2.		Mention the role of WBC as phagocyte cell.	**2**
3.		Differentiate between arteries and veins.	**2**
4.		Define heart rate and stroke volume.	**2**
5.		Define Glomerulus Filtration Rate.	**2**
6.		What is a bowman's capsule? Give its importance.	**2**
7.		Mention the role of pancreas in digestion of food.	**2**
8.		Name the hormones secreted by GI tract.	**2**
9.		Define resting potential.	**2**
10.		Enlist the components of nervous system.	**2**
11.		What do you mean by Invitro Fertilization?	**2**
12.		What is ovulation?	**2**
PART-B			
Answer any SIX questions:			**6x5=30**
13.		What are the different types of leucocytes? Give their structure with function.	**5**
14.		Describe the composition and functions of CSF.	**5**
15.		Give the structure and functions of lungs	**5**
16.		Define blood pressure and explain factors affecting blood pressure.	**5**
17.		List and explain the six major functions of kidney.	**5**
18.		How are fats digested and absorbed?	**5**
19.		Write the structure and functions of actin and myosin.	**5**
20.		Describe the principal events of oogenesis.	**5**
		PART-C	
Answer any TWO questions:			**2x10=20**
21.		Explain the formation of urine. How the acid-base balance is regulated?	**10**
22.		Describe the structure of neuron. Explain the features of the nerve impulse.	**10**
23.		Explain various types of birth control methods and compare their effectiveness	**10**

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